

ASIC RTL-to-GDSII with Open-Source Tools

Yosys/ABC synthesis, gate-level equivalence, and a pure-Python floorplan-place-route-GDSII teaching flow

AT A GLANCE

109 cells, 641 um²

counter8 gate netlist

1.64 ns, 15 levels

Critical path (mini_sta)

1.121 vs 0.753 ns

RCA vs CLA delay

2352 -> 1247 um (-47 %)

Placement HPWL

122 / 122 nets, 6 rip-ups

Routing

69.2 % (target 70 %)

Site utilisation

96 676 B, 1656

elements

GDSII stream

14 793 checks, 0

violations

DRC-lite

VERDICT

SIM PASS **GATE == RTL**

SYNTH OK **ROUTED 100 %**

GDS VERIFIED

ORFS NOT RUN

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Objective and brief mapping

The brief: synthesise an 8-bit up/down counter with load and a ripple-carry vs carry-lookahead adder pair onto the generic CMOS liberty, check gate-level equivalence against RTL, then build an educational physical-design flow in pure Python (floorplan, SA placement on HPWL, 2-layer maze routing with rip-up, DEF, a real GDSII written and re-read by my own code, DRC-lite, PNG renders), and ship an OpenROAD-flow-scripts + KLayout setup declared not run. Everything except the ORFS/KLayout run was executed here; the numbers below come from synth/, sim/ and pyflow/out/flow.log.

Flow architecture

Technology assumptions in pyflow: row height 4.0 um, routing/site pitch 0.4 um, cell width = liberty area / row height rounded up to whole tracks (one track per pin), M2 horizontal-preferred, M3 vertical-preferred, wire width 0.2 um, spacing rule 0.2 um, IO pins on the core boundary (inputs left, outputs right, clk bottom).

```
rtl/counter8.v --yosys+abc(cmos_cells.lib)--> counter8_net.v/.json --iverilog--> gate-level sim (c
|--mini_sta--> critical path / Fmax |--power
+--pyflow--> floorplan -> SA place -> Lee route
```

```
rtl/adders.v: rca4 (full_adder hierarchy kept through ABC) vs cla4 (flat, abc -D 400) -> area/dela
```

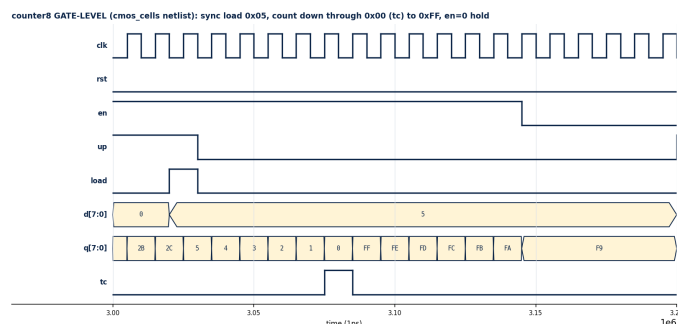
Logic synthesis and timing (Yosys 0.68 + ABC, tools/cmos_cells.lib, tools/mini_sta.py)

A flat ABC run turns both adders into the same netlist, so for a meaningful comparison rca4 kept its full_adder hierarchy through mapping (then flattened) while cla4 was mapped flat with a 400 ps delay target. The CLA delay advantage is real; its smaller area is ABC finding a better implementation than my sum-of-products carries.

Design	Cells / mix	Area um ²	Critical path	Comment
counter8	109: 8 DFF, 42 NAND, 36 NOR, 6 AND, 4 OR, 6 XNOR, 4 XOR, 3 NOT	641	1.64 ns, 15 levels (~610 MHz)	leakage 93 nW; dynamic 50.9 uW @1.8 V from gate VCD
rca4 (ripple)	24: 8 XNOR, 4 AND, 4 OR, 4 NOR, 4 NOT	156	1.121 ns, 13 levels	carry ripples through 4 mapped full adders
cla4 (lookahead)	20: 8 XOR, 12 NAND	128	0.753 ns, 9 levels	33 % faster; ABC re-factored lookahead into a NAND tree

Verification: RTL vs gate-level equivalence by simulation

Testbench	Stimulus	RTL	Gate-level	Equivalence
tb_counter8	300 up-counts with 0xFF->0x00 wrap and tc, sync load 0x05, count down through 0x00, hold, 40 random cycles, re-reset: 363 checks against a behavioural model	PASS	PASS	checksum 09434826 on both
tb_adders	exhaustive 16 x 16 x 2 = 512 vectors on rca4 and cla4 simultaneously	PASS	PASS	0 of 1024 mismatches
test_pyflow.py	GDS real8 codec, GDS write/read round-trip, malformed stream rejected, netlist counts, SA legality + improvement, HPWL, router connectivity and no shared nodes	-	-	7 / 7 OK



Gate-level netlist (cmos_cells.v models) driven by the same testbench: load 0x05, down-count through 0x00 with tc, hold on en = 0.

Physical design results (pyflow, counter8)

Step	Measured result (pyflow/out/flow.log)
Floorplan	core 34.4 x 32.0 um = 1100.8 um ² , 8 rows; target utilisation 70 % -> site utilisation 69.2 %, cell-area utilisation 58.2 % (cells snapped up to whole tracks); 22 IO pin bits on the boundary
Placement	simulated annealing, 30 000 swap/move moves, seed 1: HPWL 2352.4 um -> 1246.8 um (-47.0 %); legal: no overlaps, all cells on the 0.4 um site grid

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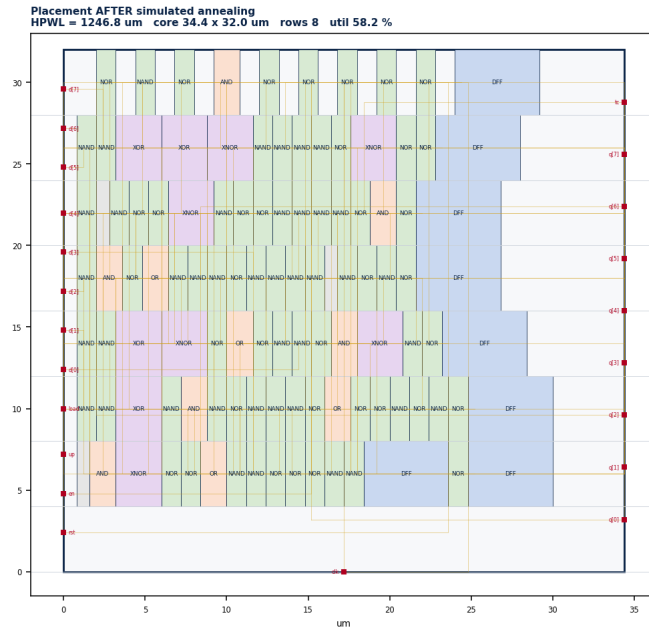
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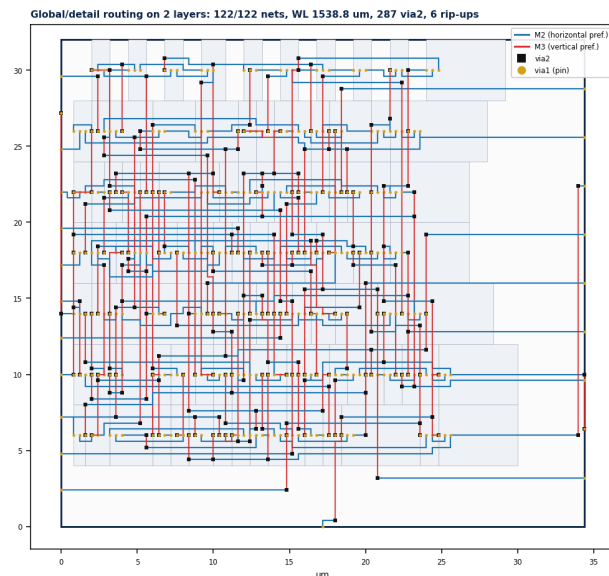
Step	Measured result (pyflow/out/flow.log)
Routing	grid 87 x 81 x 2 layers; 122 / 122 nets routed, 6 rip-ups, max 2 attempts per net; wirelength 1538.8 μm , 287 via2 + 346 via1
DEF / GDSII	counter8.def (ROWS, COMPONENTS, PINS, NETS + ROUTED); counter8.gds 96 676 bytes: 1067 BOUNDARY + 589 PATH, 1656 LAYER/DATATYPE/XY/ENDEL, 589 WIDTH, 1 each HEADER/BGNLIB/LIBNAME/UNITS/BGNSTR/STRNAME/ENDSTR/ENDLIB
GDS read-back	own reader: structure 'counter8', units 0.001 μm / 1e-9 m, element counts equal to what was written -> consistent
DRC-lite	14 793 checks: cell overlap 0, outside core 0, M2 spacing 0, M3 spacing 0, via overlap 0, zero-length wire 0 -> clean

Placement before and after annealing, routed layout



After annealing: HPWL 1246.8 μm . Gold rectangles are the per-net bounding boxes whose half-perimeters the annealer sums; the before-image (2352.4 μm) is in pyflow/out/placement_before.png.

Routed layout on two metal layers



Lee/maze routing: M2 (blue, horizontal-preferred), M3 (red, vertical-preferred), via2 (black), via1 pin accesses (gold). The clk net enters at the bottom pin and fans out to eight DFF clock pins as a normal net: no CTS in pyflow.

GDSII layer contents (from the read-back of pyflow/out/counter8.gds)

GDS layer	Purpose	Element type	Count	Geometry written
7	core boundary	BOUNDARY	1	34.4 x 32.0 μm rectangle
1	cell outlines	BOUNDARY	109	one rectangle per placed cell (width x 4.0 μm)

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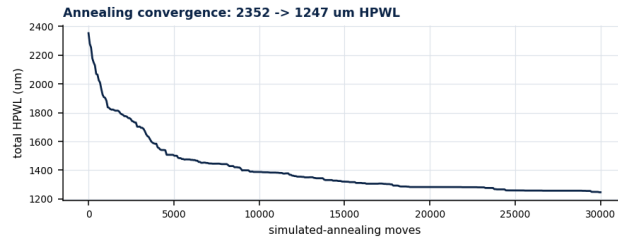
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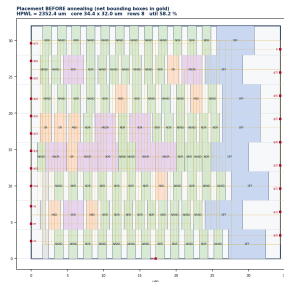
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GDS layer	Purpose	Element type	Count	Geometry written
2	M1 pins + power rails	BOUNDARY / PATH	333	324 pin stubs $0.2 \times 0.8 \mu\text{m}$ + 9 rails $0.4 \mu\text{m}$ wide (one per row edge)
3	M2 horizontal wires	PATH	418	$0.2 \mu\text{m}$ wide segments on the $0.4 \mu\text{m}$ track grid
4	M3 vertical wires	PATH	162	$0.2 \mu\text{m}$ wide segments
5	via1 (M1-M2 pin access)	BOUNDARY	346	$0.2 \times 0.2 \mu\text{m}$ squares, one per net terminal
6	via2 (M2-M3)	BOUNDARY	287	$0.2 \times 0.2 \mu\text{m}$ squares at every layer change



Simulated-annealing convergence of the total HPWL over 30 000 moves (geometric cooling, $\alpha = 0.995$ per 75-move temperature step).

Placement before annealing (round-robin row assignment)



Initial placement, HPWL 2352.4 μm : cells are dealt to rows in netlist order, so the gold net bounding boxes span the whole core; annealing shrinks them by 47 %.

Concepts vs what pyflow did

- Floorplanning: core size from area/utilisation, rows, IO pin sides -- done (no power grid beyond one M1 rail per row; ORFS adds a PDN).
- Placement: pyflow keeps cells in ordered row lists so every state is legal and anneals on HPWL; OpenROAD uses analytical global placement (RePIAce) plus detailed placement.
- CTS: not done in pyflow (clk routed as a 9-terminal net); ORFS runs TritonCTS after placement.
- Routing: 2-layer track grid, weighted Lee expansion (wrong-way x3, via x4), incremental multi-terminal trees, rip-up and re-route with a history penalty; ORFS uses FastRoute + TritonRoute with real rules.
- DRC/LVS: DRC-lite (spacing, overlap, containment) plus a connectivity check equivalent to LVS at net level; real DRC/LVS need Magic/KLayout with PDK decks.
- GDSII: written record by record (big-endian lengths, excess-64 base-16 8-byte reals) and re-parsed to prove the structure.

Declared substitutions / not run here

- OpenROAD-flow-scripts, sky130 PDK, KLayout, Magic, OpenSTA are not in the container and were not run. orfs/config.mk (sky130hd, $60 \times 60 \mu\text{m}$ die, 10 ns clock, met1-met4), orfs/constraint.sdc, orfs/run_orfs.sh (native make and Docker: `docker run --rm -it -v $(pwd)/flow:/OpenROAD-flow-scripts/flow -w /OpenROAD-flow-scripts/flow openroad/orfs:latest make DESIGN_CONFIG=designs/sky130hd/counter8/config.mk`) and two KLayout .lyp files with orfs/view.py are ready to use.
- The GDS made here is a structurally valid stream in the educational technology of tools/cmos_cells.lib (row $4 \mu\text{m}$, pitch $0.4 \mu\text{m}$), not a manufacturable sky130 layout: no diffusion/poly/contact layers inside cells, no PDN, no fill, no CTS.
- Timing from tools/mini_sta.py (teaching-grade RC model), power from tools/power_estimate.py; simulation with Icarus Verilog.

Learnings / next steps

The first router run failed 36 of 122 nets: pin offsets on half-tracks collapsed onto one grid node under round-half-even. Whole-track pins, pin nodes as hard obstacles and a negotiated-congestion history cost brought it to 122/122 with 6 rip-ups. Next: timing-driven net weights from mini_sta, an H-tree clock, an M1 pin-access model, and running the shipped ORFS configuration on my machine to compare sky130hd results.